

Hierarchical RemoteCLIP Distillation for Channel-Robust Semantic Communication of Remote Sensing Imagery

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Abstract—Remote sensing (RS) downlinks deployed on UAV inspection, emergency response, and Earth-observation small-satellite platforms operate under low signal-to-noise ratio (SNR), fading, and limited link margin. Conventional “compress-then-channel-code” pipelines exhibit cliff-effect collapse at low SNR, while rigid full-image transmission wastes scarce bandwidth even when only a coarse class label is needed at the receiver. We present RS-Token, a hierarchical discrete tokenizer for remote-sensing image downlinks. A four-stage residual vector quantizer (RVQ) encodes the image into a multi-layer index tensor; RemoteCLIP distillation forces the first codebook layer (L0) to carry land-cover semantic identity, while the remaining layers encode pixel-level detail. The receiver adaptively selects the first k index layers based on channel quality and downstream task requirement, transmitting only L0 (2,560 bits/img) to preserve task fidelity under poor channels and stacking subsequent layers (up to 10,240 bits/img) when the link improves. On the AID 30-class scene-classification protocol with BPSK over AWGN/Rayleigh-fading channels, distillation lifts the linear-probe accuracy of L0 indices from 57.7% to 82.4% (+24.7 pp), with the first layer alone covering 95% of the distillation-induced gain. In the $\text{SNR} \geq +5$ dB regime, transmitting only L0 yields 25–32 pp higher classification accuracy than the no-distillation 4-layer transmission while using $4\times$ less bandwidth. A teacher ablation shows that RemoteCLIP achieves a stable 4–7 pp advantage over a generic OpenAI CLIP teacher in deployment-relevant mid-SNR channels, demonstrating the role of remote-sensing-specific pretraining in channel-robust task fidelity.

Index Terms—Semantic communication, remote sensing, residual vector quantization, foundation-model distillation, RemoteCLIP, graceful degradation, task-oriented communication.

I. INTRODUCTION

REMOTE-SENSING downlinks deployed on UAV inspection, disaster response, edge sensor networks, and CubeSat/SmallSat platforms experience large SNR fluctuations driven by distance, occlusion, Doppler shift, interference, and amateur-band operation. Receivers in these scenarios are commonly required to perform real-time tasks such as land-cover recognition under unstable channel conditions. The information demand of these tasks varies widely across abstraction levels: automatic alerts only need coarse class labels, emergency dispatch requires the spatial extent of affected regions and damaged structures, while subsequent mapping and archival demand high-fidelity reconstruction.

Conventional digital transmission paradigms struggle to span both ends of this demand. “Compress-then-channel-code”

pipelines (e.g., JPEG2000+LDPC, BPG+LDPC) exhibit cliff-effect collapse below a channel-quality threshold and lose task capability entirely; rigid full-image transmission, on the other hand, occupies the full bit budget even when the receiver needs only a coarse label, wasting scarce link margin. These limitations jointly motivate a working regime not fully covered by existing schemes: *the receiver must preserve downstream task accuracy with as few codebook indices as possible while retaining the option to refine pixel quality on demand when the link improves*. RS-Token operates above this application-layer demand, orthogonal to and stackable with physical-layer mechanisms such as Hybrid ARQ (HARQ) and adaptive coding and modulation (ACM); their relationship is detailed in Section V-B.

A natural design hypothesis is to dedicate the first layer of a discrete codebook to the most task-critical information—land-cover identity in the RS context—while the remaining layers progressively refine pixel detail. The receiver then selects the first k layers according to channel and task, transmitting L0 alone under poor channels and stacking subsequent layers when the link is favorable. This pattern has matured in the audio domain: SpeechTokenizer [1] distills RVQ layer 1 from HuBERT to encode phonemic content while the residual layers carry acoustic detail; STACodec, DM-Codec, HAC and LM-SPT extend the same principle. No counterpart yet exists in the visual or remote-sensing domain.

In digital semantic communication, MOC-RVQ [2] and ReVQom [3] use RVQ for multi-layer discrete-token transmission but focus on natural images or V2X cooperative perception, treat all RVQ layers as residual-precision refinements, leave the first layer without semantic identity, and do not distill any foundation model. MSVQ-SC and ESC-MVQ refine multi-codebook joint modulation–power optimization without introducing foundation-model supervision. On the visual representation side, BEiT v2 [4] distills a single-layer VQ codebook from CLIP/DINO for masked image modeling; VILA-U [5] supervises an entire residual quantizer for multimodal LLMs without hierarchical decoupling and without channel evaluation; SemCLIP [6] transmits CLIP tokens for semantic communication using a generic OpenAI CLIP teacher and a single-layer codebook. In task-aware RS transmission, Gao *et al.* [7] apply DRL block selection with block compressed sensing to AID classification under Rayleigh+AWGN—using exactly the protocol followed in this work—but without codebooks, RVQ, or foundation-model distillation; the Luxembourg Snt EO+JSCC family [8] pursues

continuous JSCC rather than discrete RVQ; MAGC uses VAE+diffusion reconstruction for low-rate RS compression and does not enter the codebook-index transmission framework.

To fill this gap, we propose **RS-Token**, a hierarchical RVQ tokenizer for RS downlinks distilled by RemoteCLIP [9]. A cosine-alignment loss supervises the first RVQ codebook to carry land-cover semantic identity while the remaining three layers carry pixel detail; the receiver adaptively selects the first k layers ($k \in \{1, 2, 3, 4\}$) under BPSK over AWGN/Rayleigh fading. Under the AID 30-class classification protocol the contributions are fourfold. **(i)** Distillation raises L0-index linear-probe accuracy from 57.7% to 82.4% (+24.7 pp) at a reconstruction PSNR cost of only 0.22 dB—well below the 0.5 dB design budget. **(ii)** Transmitting only L0 (2,560 bits/img) with distillation outperforms the no-distillation full 4-layer transmission (10,240 bits/img) by 25 pp in classification accuracy across the SNR $\geq +5$ dB regime, achieving $4\times$ bandwidth reduction with concurrent accuracy gain. **(iii)** At -5 dB AWGN, $k = 1$ (8.2%) still outperforms $k = 4$ (5.7%); graceful degradation holds in the same direction under both AWGN and Rayleigh fading, on the protocol matched to Gao *et al.* [7]. **(iv)** A teacher ablation shows RemoteCLIP achieving a 4–7 pp task-fidelity advantage over OpenAI CLIP in deployment-relevant mid-SNR channels (AWGN 0 dB; Rayleigh +5 to +10 dB), supported by direct geometric measurements on the L0 codebook.

II. RELATED WORK

A. Multi-layer discrete tokens for digital semantic communication

VQ-VAE [10] and Residual VQ [11] jointly establish the methodological basis for discrete visual tokens and multi-stage quantization, the latter further popularized in the audio domain by SoundStream [12] and EnCodec [13] for generative codecs. MOC-RVQ [2] aligns multi-head codebook indices to digital constellations (64-QAM) and validates “index-on-modulation” end-to-end on natural images. ReVQom [3] pushes RVQ multi-stage compression to $273\text{--}1365\times$ ratios in V2X multi-agent perception, but assumes error-free transmission and treats layers as residual precision. The MSVQ-SC and ESC-MVQ family integrates multi-codebook design with joint modulation–power optimization and rate adaptation, yet preserves a reconstruction-and-channel-geometry objective without foundation-model supervision. The common signature of these schemes is to leave RVQ layers as residual-precision refinements without semantic specialization, evaluating on PSNR/LPIPS rather than RS task fidelity.

B. Foundation-model distillation into discrete visual tokens

BEiT v2 [4] distills a single-layer VQ codebook from CLIP/DINO for masked image modeling rather than communication. VILA-U [5] supervises an entire residual quantizer with a CLIP text-alignment loss, mapping the full chain to a unified representation rather than the SpeechTokenizer-style “layer-1 for semantics, residual layers for detail” hierarchy, and is not evaluated under channel simulation. TokLIP, UniTok

and MUSE-VL further advance semantic visual tokenizers via joint training or dual-path architectures, again targeted at LLMs and multimodal generation rather than channel-aware transmission. SemCLIP [6] comes closest to communication settings by transmitting CLIP tokens directly, but uses generic OpenAI CLIP and retains a single-layer codebook. The common gap is the absence of hierarchical decoupling, almost no channel-simulation evaluation, and no remote-sensing-pretrained teacher.

C. Task-aware transmission in remote sensing

Gao *et al.* [7] apply DRL block selection with 4×4 block compressed sensing, 8-bit quantization, and ResNet34 classification on AID under Rayleigh+AWGN—using exactly the protocol of this paper—but with a 2022-vintage stack lacking codebooks, RVQ, and foundation-model distillation. The Luxembourg SnT team consistently produces EO+JSCC work (including LEO-JSCC-EO at ICC 2026), pursuing continuous JSCC rather than discrete RVQ; the foundational DeepJSCC [8] anchors this line. MAGC employs VAE+diffusion reconstruction with vector-map conditioning for low-rate RS compression, outside the codebook-index transmission framework. FOOL and related work address task-agnostic RS compression without codebook indices. Collectively these works span RS task-fidelity protocols and digital/continuous JSCC, but the combination “discrete codebook + multi-stage RVQ + RS-pretrained foundation-model distillation” has not previously appeared.

D. Audio-domain analogue

SpeechTokenizer [1] supervises RVQ layer 1 with HuBERT layer 9 to encode phonemes while residual layers carry timbre and prosody, establishing the “hierarchical distilled tokenizer” paradigm in the audio domain; STACodec, DM-Codec, HAC, and LM-SPT extend the same mechanism. Stable operation in this closely related modality reduces the scientific risk of transferring the mechanism to vision and remote sensing.

III. METHOD

A. Framework and Notation

Overall architecture. An input image $x \in \mathbb{R}^{3\times 256\times 256}$ is mapped by an encoder E to a feature map $z \in \mathbb{R}^{256\times 16\times 16}$. After spatial flattening, the feature map yields $T = 256$ patch tokens of dimension $d = 256$. This token sequence is processed by a 4-stage Residual VQ quantizer with codebook size $K = 1024$ per layer. Denoting the ℓ -th codebook as $\mathbf{C}^{(\ell)} \in \mathbb{R}^{K\times d}$, quantization yields an index tensor $\mathbf{I} \in \{0, \dots, K-1\}^{T\times 4}$. Each index occupies $\log_2 K = 10$ bits; the per-image bit budget is $T \cdot 4 \cdot \log_2 K = 256 \cdot 4 \cdot 10 = 10,240$ bits for full 4-layer transmission, and $T \cdot 1 \cdot \log_2 K = 2,560$ bits for L0-only transmission.¹ The receiver reconstructs $\hat{z}_q$ from the first k received layers ($k \in \{1, 2, 3, 4\}$) as $\hat{z}_q = \sum_{\ell=1}^k \mathbf{C}^{(\ell)}[\hat{\mathbf{I}}_\ell]$, then a decoder D produces the reconstructed

¹The budget excludes the receiver-side signaling field that announces the value of k . Encoding $k \in \{1, 2, 3, 4\}$ requires only 2 bits per image, negligible relative to the 2,560–10,240-bit per-image transmission.

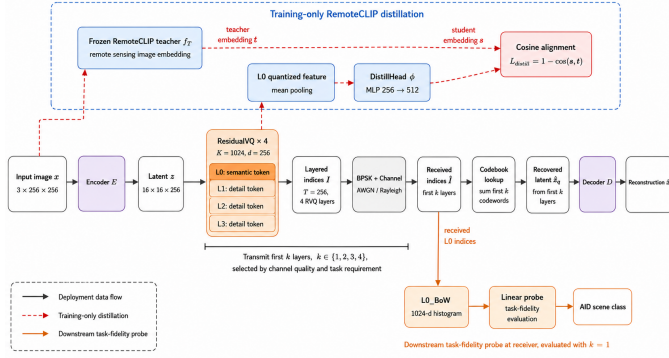


Fig. 1. RS-Token architecture. **Transmitter side (left of dotted divider):** an encoder maps the input image x to a 16×16 spatial latent z , quantized by a 4-stage Residual VQ ($K = 1024$, $d = 256$) into the index tensor $\mathbf{I}$. The first k layers are transmitted via BPSK over AWGN/Rayleigh channels. **Receiver side (right of divider):** the received indices $\hat{\mathbf{I}}$ feed two paths—a reconstruction path through codebook lookup and decoder D that produces $\hat{x}$, and a task path (deployment, $k = 1$) where the L0 index histogram h_0 feeds a linear classifier. **RemoteCLIP distillation (top, training only):** the L0 quantized embedding is mean-pooled and projected through DistillHead ϕ to align with the frozen RemoteCLIP teacher f_T via a cosine loss $\mathcal{L}_{\text{distill}} = 1 - \cos(s, t)$. The choice of k is dictated jointly by channel quality and task requirement, enabling task-fidelity-preserving graceful degradation.

image $\hat{x}$. The task-output path constructs a 1024-dimensional histogram feature h_0 (Section III-E) directly from the received L0 indices and feeds it to a single-layer logistic regression to produce the task label.

Encoder/decoder backbone. Encoder E and decoder D follow the VQ-VAE-2 / VQ-GAN-style convolutional residual backbone. The encoder consists of four downsampling stages with channel multipliers $[1, 2, 4, 4] \times C_{\text{base}}$, $C_{\text{base}} = 64$. Each stage contains one residual block (GroupNorm + SiLU + 3×3 convolutions) and one stride-2 downsampling convolution, achieving $16 \times$ spatial downsampling with the channel dimension expanded to $d = 256$. The decoder mirrors the encoder using stride-2 transposed convolutions for upsampling. The full encoder + decoder + RVQ + DistillHead stack contains approximately 10.87M parameters (encoder 5.16M, decoder 5.71M, quantizer + DistillHead < 0.5 M). All intermediate layers use GroupNorm and SiLU activations; self-attention is not used so that complexity remains tractable for a short-form architecture.

B. RemoteCLIP Distillation

The L0 quantized output $\mathbf{z}_q^{(1)} \in \mathbb{R}^{T \times d}$ is mean-pooled across patch tokens to yield $\bar{\mathbf{z}} \in \mathbb{R}^d$, then projected through a 2-layer MLP head ϕ to the RemoteCLIP image-embedding space, producing $\mathbf{s} = \phi(\bar{\mathbf{z}}) \in \mathbb{R}^{512}$. The teacher path feeds the original image x into the frozen RemoteCLIP visual encoder f_T to obtain $\mathbf{t} = f_T(x) \in \mathbb{R}^{512}$. The distillation loss is the cosine alignment

$$\mathcal{L}_{\text{distill}} = 1 - \cos(s, t). \quad (1)$$

The full distillation path is highlighted by the red dashed lines along the top of Fig. 1 and is active during training only.

RemoteCLIP [9] is pretrained on 1.65M remote-sensing image-text pairs and reaches 95.95% linear-probe accuracy

on AID, exceeding the ImageNet-ViT-B baseline of 83.55% by 12.4 pp—a clear remote-sensing domain inductive bias. In the feature space this bias manifests as the 30 AID class clusters maintaining larger inter-cluster spacing and tighter intra-cluster compactness; Section IV-F reports the downstream task-fidelity consequence of transferring this geometric property to the L0 codebook through distillation.

C. Total Loss

The total loss combines pixel reconstruction, perceptual loss, vector-quantization commitment loss, and distillation loss:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + 0.1 \cdot \mathcal{L}_{\text{LPIPS}} + 0.25 \cdot \mathcal{L}_{\text{VQ}} + \lambda \cdot \mathcal{L}_{\text{distill}}. \quad (2)$$

Here $\mathcal{L}_{\text{recon}}$ is the L1 reconstruction loss, $\mathcal{L}_{\text{LPIPS}}$ is the AlexNet-backed LPIPS [14], and $\mathcal{L}_{\text{VQ}}$ is the standard Residual VQ commitment loss. The distillation weight λ is set to 0.5 following the trade-off sweep in Section IV-G, balancing in-domain task accuracy and noise robustness.

D. Channel Simulation Protocol

The transmitter pipeline maps each index to its 10-bit binary representation followed by BPSK modulation. Two channel models are evaluated. Under AWGN, the closed-form BPSK hard-decision bit-error rate is $\text{BER} = \frac{1}{2} \text{erfc}(\sqrt{\text{SNR}_{\text{lin}}})$. Under Rayleigh flat fading combined with AWGN, averaging the instantaneous BER over $|h|^2 \sim \text{Exp}(1)$ yields $\text{BER} = \frac{1}{2} (1 - \sqrt{\text{SNR}_{\text{lin}} / (1 + \text{SNR}_{\text{lin}})})$. Both expressions are standard BPSK results for the respective channel models [15]. The simulation flips each transmitted bit with the corresponding closed-form BER probability and reassembles the receiver-side indices, equivalent to the Monte Carlo average of the full waveform-level “BPSK $\rightarrow$ AWGN $\rightarrow$ hard decision” pipeline. We cross-validated the closed-form approach against waveform-level Monte Carlo simulation at ± 5 dB and ± 10 dB on 10^6 -bit batches; the two approaches agree statistically.

E. Receiver Task-Fidelity Feature h_0

When only the L0 layer is transmitted ($k = 1$), the receiver constructs a 1024-dimensional histogram feature directly from the received index sequence without any codebook lookup:

$$h_0[i] = \frac{1}{T} \sum_{t=1}^T \mathbb{1}[\hat{\mathbf{I}}_t^{(1)} = i], \quad i = 0, \dots, K-1. \quad (3)$$

The feature h_0 depends only on the statistical pattern of the received indices, not on codebook geometry or the decoder; it is the most parsimonious information form available to the receiver under L0-only transmission. After standardization (column-wise division by standard deviation, without mean centering, to preserve sparse-frequency structure) the feature feeds a single-layer logistic regression (LBFGS solver, $\text{max_iter} = 2000$, $C = 1.0$) for task-label output. The task path is shown along the bottom branch of Fig. 1. The argument here is that if such a minimalist representation already attains 80%+ accuracy on AID 30-class classification, then the L0 indices encode land-cover identity directly through their discrete combinatorial pattern, requiring neither downstream decoding nor geometric information.

TABLE I
TRAINING-STAGE FINAL VALUES AND LINEAR-PROBE COMPARISON (AID TEST, NO-CHANNEL TRANSMISSION, SINGLE-SEED RUN).

Configuration	PSNR (dB)	LPIPS	h_0 acc. (%)
Stage 1 (single VQ)	23.80	0.242	—
Stage 2 (RVQ $\times$ 4, no distill.)	26.10	0.172	57.7
Stage 3 (RVQ$\times$4 + RemoteCLIP)	25.88	0.176	82.4

IV. EXPERIMENTS

A. Dataset and Protocol

Experiments use AID (Aerial Image Dataset, 30-class scene classification) [16]. Training/validation/test splits sample 8,000 / 1,000 / 1,000 images at an 8:1:1 ratio. The evaluation protocol structure matches the UAV task-aware transmission protocol of Gao *et al.* [7] (AID 30-class + classification accuracy + AWGN/Rayleigh channel sweep). The specific split list is generated internally with random seed 42; the full list with per-class counts is included in the supplementary material. All images are resized to 256×256 RGB and normalized to $[-1, 1]$.

B. Training Configuration

The model uses 4-stage Residual VQ with codebook size 1,024 and latent dimension 256. Training spans 50 epochs at batch size 16, learning rate 1×10^{-4} , AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.95$), 500-step linear warmup, gradient clipping 1.0, bf16 mixed precision, random seed 42. The teacher network RemoteCLIP-ViT-B/32 remains frozen throughout. Training runs on a single NVIDIA RTX 5070 Ti card; one full training session takes approximately 70 minutes. No codebook collapse is observed: all four codebook layers maintain $> 99\%$ utilization throughout training (measured as the ratio of unique codeword hits per full epoch to K).

C. Main Result: Distillation Lifts L0 Task Fidelity

Table I summarizes training-stage final values and L0 linear-probe accuracy on the AID test set. Stage 1 (single VQ) lacks the multi-layer structure required for h_0 /L0-emb/zq-pool features and is marked “—”.

PSNR and LPIPS report the best validation values across 50 training epochs. Stage 3 incurs only a 0.22 dB PSNR cost relative to Stage 2, well below the 0.5 dB design budget. The L0 index histogram h_0 classification accuracy rises from 57.7% to 82.4%, a +24.7 pp gain that is nearly five times the preset 5 pp threshold.

D. Channel Simulation: Task-Fidelity Graceful Degradation

Fig. 2 displays the $\text{SNR} \times k$ accuracy matrix of RS-Token (Stage 3) under AWGN and Rayleigh fading channels (the two panels share a colorbar for direct comparison). Table II extracts the most diagnostic comparison: “distilled + L0-only (2,560 bits/img)” versus “no-distillation full 4-layer (10,240 bits/img)”.

In the $\text{SNR} \geq 0$ dB regime, transmitting only L0 (a $4\times$ reduction in bit rate) yields 25 to 32 pp higher classification

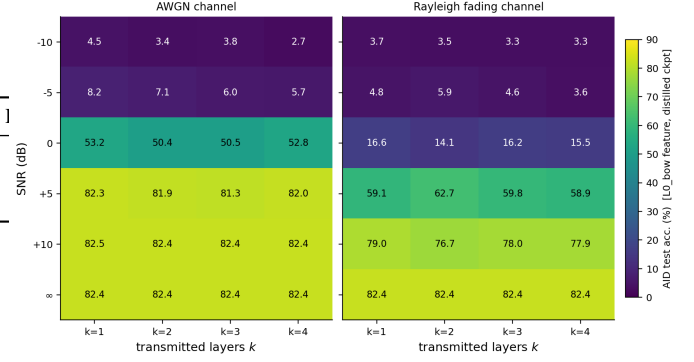


Fig. 2. Task-fidelity degradation under AWGN and Rayleigh fading channels. Each cell reports the AID test classification accuracy of the distilled model at the corresponding SNR and number of transmitted layers k , computed from h_0 features.

TABLE II
 h_0 CLASSIFICATION ACCURACY (AID TEST, SINGLE-SEED RUN):
DISTILLED $k=1$ VERSUS NO-DISTILLATION $k=4$.

SNR	Distilled $k=1$ (2,560 bits)	No-distill. $k=4$ (10,240 bits)	Δ (pp)
0 dB AWGN	53.2	21.2	+32.0
+5 dB AWGN	82.3	57.2	+25.1
+10 dB AWGN	82.5	57.6	+24.9
+5 dB Rayleigh	59.1	27.2	+31.9
+10 dB Rayleigh	79.0	48.2	+30.8

accuracy than the no-distillation 4-layer transmission, indicating that the semantic information density carried by L0 indices after distillation substantially exceeds the contribution of full multi-layer reconstruction to classification. At -5 dB AWGN, $k = 1$ (8.2%) still outperforms $k = 4$ (5.7%), although both approach random performance ($1/30 \approx 3.3\%$); at Rayleigh 0 dB, $k = 1$ further drops to 16.6%, marking a clear breakdown point where physical-layer techniques such as LDPC, HARQ, or equalization are needed to restore the link (Section V-B). We therefore restrict the graceful-degradation claim to the $\text{SNR} \geq +5$ dB deployment regime, reporting Rayleigh 0 dB and lower-SNR behavior as the breakdown boundary.

E. Layer Specialization: L0 is the Semantic Layer

Fig. 3 compares cumulative-layer linear-probe accuracy under no-distillation and distilled configurations.

Under distillation, the total increment from L_0 (86.0%) to $L_0 + L_1 + L_2 + L_3$ (88.0%) is only +2.0 pp—i.e., the first layer alone accounts for $86.0/88.0 = 97.7\%$ of the full 4-layer accuracy and $(86.0 - 47.7)/(88.0 - 47.7) = 95.0\%$ of the distillation-induced gain over the no-distillation L0. Without distillation, all four cumulative-layer accuracies remain at 47–48%, indicating that RVQ does not spontaneously develop a task-relevant semantic hierarchy without external supervision; distillation is the principal driver of L0 semanticization. This division of labor mirrors the SpeechTokenizer observation in audio that “the first layer carries content while subsequent layers carry detail”, here quantified for the visual / remote-sensing domain.

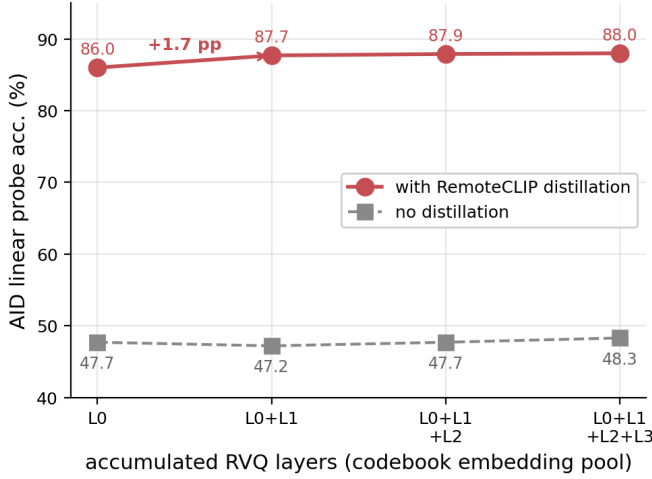


Fig. 3. Cumulative-layer linear-probe accuracy on AID test top-1 under no-channel transmission. RemoteCLIP distillation makes L0 alone capture the principal semantic discriminative power, whereas RVQ without distillation forms no task-relevant semantic hierarchy across cumulative layers.

TABLE III

TEACHER ABLATION h_0 CLASSIFICATION ACCURACY (AID TEST, $k=1$, SINGLE-SEED RUN, %).

Setting	OpenAI CLIP	RemoteCLIP	Δ (pp)
No channel (in-domain ceiling)	80.8	82.4	+1.6
AWGN -5 dB	6.2	8.2	+2.0
AWGN 0 dB	46.5	53.2	+6.7
AWGN +5 dB	79.4	82.3	+2.9
AWGN +10 dB	80.8	82.5	+1.7
Rayleigh +5 dB	55.3	59.1	+3.8
Rayleigh +10 dB	74.8	79.0	+4.2

F. Teacher Ablation: RS-Specific Foundation Model in Channel Downstream

Table III compares RemoteCLIP and OpenAI CLIP as distillation teachers. The two configurations only differ in the teacher weights; architecture, hyperparameters, and training pipeline are otherwise identical.

The teacher gap is only 1.6 pp at the no-channel ceiling, but enlarges to 4–7 pp in the deployment-relevant mid-SNR regime (AWGN 0 dB; Rayleigh +5 to +10 dB). Compared with OpenAI CLIP’s 400M natural-image-text pretraining, RemoteCLIP’s 1.65M remote-sensing image-text pretraining provides a markedly stronger geometric prior for remote-sensing scenes; distillation transfers this prior to the L0 codebook, yielding more stable downstream task fidelity in mid-SNR channels.

To quantify the geometric mechanism above, we extracted the L0 codebook embedding (8,000 images $\times$ 256-dim mean-pooled features) from each teacher’s distilled checkpoint on the AID training set and computed cluster-structure metrics over the 30 classes (Table IV). RemoteCLIP outperforms OpenAI CLIP consistently across all four metrics: inter/intra-cluster distance ratio 1.82 vs. 1.72; silhouette coefficient 0.059 vs. -0.004 (the latter approaches random distribution); Calinski-Harabasz index 404 vs. 366; Davies-Bouldin index 2.32 vs. 2.83 (lower is better). The most informative gap is

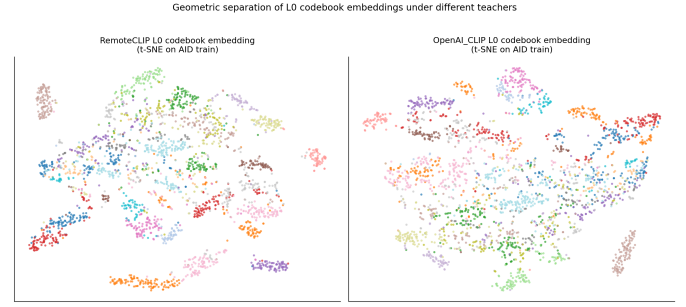


Fig. 4. t-SNE visualization of L0 codebook embeddings under the two teachers (AID training images, 2,500 sampled, 30 classes color-coded). RemoteCLIP (left) shows clearer cluster separation and tighter intra-cluster structure, consistent with the quantitative metrics in Table IV.

TABLE IV

L0 CODEBOOK CLUSTER-STRUCTURE METRICS (AID TRAINING SET $N = 8,000$, 30 CLASSES, SINGLE-SEED RUN).

Metric	OpenAI CLIP	RemoteCLIP	Direction
Inter/intra distance ratio	1.7231	1.8156	$\uparrow$ higher better
Silhouette score	-0.0039	0.0592	$\uparrow$ higher better
Calinski-Harabasz score	366.03	404.06	$\uparrow$ higher better
Davies-Bouldin score	2.8326	2.3162	$\downarrow$ lower better

TABLE V

DISTILLATION-WEIGHT SWEEP (AID TEST, $k=1$, h_0 , IN %; PSNR IN DB).

λ	PSNR	h_0 no-channel	h_0 AWGN 0 dB	h_0 Rayleigh +5 dB
0.0	26.10	57.7	23.4	28.0
0.1	26.17	71.2	35.5	42.1
0.5	25.88	82.4	53.2	59.1
1.0	25.61	84.5	37.5	49.8

in the silhouette coefficient: OpenAI-CLIP-distilled codebooks barely form effective cluster structure on AID’s 30 classes, whereas RemoteCLIP yields visibly separable clusters. Fig. 4 provides the matching t-SNE visualization of the L0 codebook embeddings on AID training images under the two teachers.

The 4–7 pp task-fidelity advantage of RemoteCLIP in the most deployment-relevant mid-SNR regime is supported directly at the codebook geometry level by Table IV, identifying it as the principal value of remote-sensing-specific pretraining for RS semantic communication, rather than a small in-domain ceiling improvement.

G. Distillation-Weight Trade-off and λ Selection

Table V reports the distillation weights $\lambda \in \{0.0, 0.1, 0.5, 1.0\}$ across reconstruction PSNR, in-domain classification accuracy, and channel-downstream classification accuracy.

$\lambda = 1.0$ surpasses $\lambda = 0.5$ at the in-domain ceiling by 2.1 pp, but is 15.7 pp lower than $\lambda = 0.5$ in the AWGN 0 dB regime. One interpretation is that strong distillation overly compresses codebook geometry, reducing inter-cluster spacing so that single-bit flips more easily cross class boundaries, undermining noise robustness. $\lambda = 0.5$ represents the frontier balance among reconstruction quality, in-domain classification, and channel-downstream task fidelity; choosing this weight

reflects a deployment-driven design principle—task fidelity over single-metric validation accuracy.

V. DISCUSSION

A. Robustness Amplification of Remote-Sensing Pretraining under Channel

The teacher gap is only 1.6pp at the no-channel ceiling but widens to 4–7pp in the AWGN 0dB and Rayleigh +5 to +10dB deployment regimes (Section IV-F). A consistent geometric explanation is the following: OpenAI CLIP’s feature space, learned from 400M natural-image-text pairs, “flattens” remote-sensing data only as a byproduct of cross-domain transfer; RemoteCLIP’s targeted pretraining on remote-sensing image-text pairs supplies a stronger domain-specific geometric prior, expressed as larger inter-cluster spacing and tighter intra-cluster compactness for the 30 AID classes. Table IV and Fig. 4 directly quantify this difference: the L0 codebook from RemoteCLIP distillation outperforms OpenAI CLIP across all four cluster-structure metrics; the silhouette coefficient transitions from -0.004 (near-random) to 0.059 (clearly positive separation), and the Davies-Bouldin score drops from 2.83 to 2.32 ($\sim 18\%$ improvement in cluster compactness); the t-SNE visualization shows the same direction of difference. Distillation transfers this geometric property to the L0 codebook, reducing the probability that bit-error perturbation crosses class boundaries. This mechanism complements the observation in Section IV-G that “excessive compactness undermines robustness”: remote-sensing-specific pretraining does not produce a more compact codebook, but rather more stable cluster centroids while preserving inter-cluster spacing.

B. Relation to Physical-Layer HARQ and ACM

The hierarchical degradation mechanism does not replace physical-layer HARQ and ACM; the two operate at different layers and stack additively. HARQ reduces bit error rate via retransmission at the cost of latency and uplink bandwidth, not always tolerable in time-critical scenarios such as UAV real-time backhaul or emergency dispatch. ACM has a coding-rate floor: when SNR drops below the lowest threshold, the link cannot be established. Physical-layer “error-free transmission” does not equate to application-layer “task usability”: an image transmitted with BER 10^{-6} may still fail downstream classification due to extremely low source-coding rate and blurry reconstruction. RS-Token operates at the application layer above the physical layer, providing task-fidelity fallback: when physical-layer techniques are exhausted, the receiver can still preserve task accuracy with as few codebook indices as possible. The Rayleigh 0dB measurement ($k = 1$ accuracy 16.6%) corroborates this positioning—severe fading regimes still require LDPC, HARQ, and equalization; our method is one layer above, not a replacement.

C. Limitations and Future Work

(L1) This short paper does not yet conduct end-to-end same-SNR-sweep comparisons against external baselines such as JPEG2000+LDPC, DeepJSCC, or Gao *et al.* [7]. The full

version will start from the protocol-matched Gao *et al.* 2022 baseline and progressively extend to head-to-head comparisons with classical and neural baselines in a journal version.

(L2) The current experiments are confined to RGB three-band imagery and use a single training seed. The full version will report mean $\pm$ std across 3 model seeds and 5 channel seeds and extend to multispectral / hyperspectral remote-sensing data.

(L3) A clear breakdown point appears at Rayleigh 0dB ($k = 1$ at 16.6%); subsequent work could introduce lightweight channel coding on the codebook indices (e.g., (15,11) BCH or short-length LDPC) to further tighten the task floor while preserving the overall task-fidelity hierarchical-degradation framework.

(L4) Current experiments use BPSK modulation. The framework is naturally compatible with higher-order modulations (16-QAM, 64-QAM) and corresponding BER curves; the BPSK choice corresponds to the worst-case link conditions of interest in the motivation (low SNR / amateur band), with empirical extension to higher-order modulation deferred to future work.

VI. CONCLUSION

We have presented RS-Token, a hierarchical distilled RVQ tokenizer for remote-sensing image downlinks. RemoteCLIP distillation drives the first RVQ codebook to carry land-cover semantic identity while subsequent layers carry pixel detail; the receiver dynamically selects the first k index layers based on channel quality and task requirement, achieving task-fidelity-preserving and bandwidth-adaptive transmission across the deployment SNR regime. The framework provides a new task-oriented semantic-communication paradigm for real-world remote-sensing scenarios such as UAV, emergency response, edge sensors, and SmallSats. Subsequent work will extend external-baseline comparison, multi-seed statistics, lightweight channel coding, and multispectral scenarios.

REFERENCES

- [1] X. Zhang, D. Zhang, S. Li, Y. Zhou, and X. Qiu, “SpeechTokenizer: Unified speech tokenizer for speech language models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2024.
- [2] A. list TODO], “MOC-RVQ: Multi-head octonary codebook residual vector quantization for image semantic communication,” in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2024, tODO: confirm authors and full title.
- [3] —, “ReVQom: Residual vector quantization for v2x cooperative perception,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, tODO: confirm authors and full title.
- [4] Z. Peng, L. Dong, H. Bao, Q. Ye, and F. Wei, “BEiT v2: Masked image modeling with vector-quantized visual tokenizers,” *arXiv preprint arXiv:2208.06366*, 2022.
- [5] V.-U. authors TODO], “VILA-U: a unified foundation model integrating visual understanding and generation,” in *Proc. International Conference on Learning Representations (ICLR)*, 2025.
- [6] S. authors TODO], “SemCLIP: A CLIP-based approach to semantic communication,” *arXiv preprint arXiv:2502.18200*, 2025.
- [7] Z. Gao and [coauthors TODO], “Task-oriented semantic communication for UAV remote sensing,” *IEEE Transactions on Geoscience and Remote Sensing*, 2022, tODO: confirm coauthors / volume / pages.
- [8] E. Boursoulatz, D. B. Kurka, and D. Gündüz, “Deep joint source-channel coding for wireless image transmission,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 3, pp. 567–579, 2019.

- [9] F. Liu, D. Chen, Z. Guan, X. Zhou, J. Zhu, Q. Ye, L. Fu, and J. Zhou, "Remoteclip: A vision language foundation model for remote sensing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.
- [10] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [11] D. Lee, C. Kim, S. Kim, M. Kim, and W.-S. Han, "Autoregressive image generation using residual quantization," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11 523–11 532.
- [12] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "SoundStream: An end-to-end neural audio codec," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.
- [13] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *arXiv preprint arXiv:2210.13438*, 2022.
- [14] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 586–595.
- [15] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. McGraw-Hill, 2007.
- [16] G.-S. Xia, J. Hu, F. Hu, B. Shi, X. Bai, Y. Zhong, L. Zhang, and X. Lu, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 7, pp. 3965–3981, 2017.